Prime-Factor Formulas and a Primality Criterion for Euler's Totient

A note on OEIS A000010

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Abstract

For the Euler totient sequence $\varphi(n)$ (OEIS A000010), we give self-contained elementary proofs of its prime-factor formula, divisor-sum identity, parity property, and a precise characterization of prime inputs. All results are standard and independently checkable; none is claimed to be new.

1 Definition and prime-factor formula

For $n \geq 1$, set $\varphi(n) = \#\{1 \leq a \leq n : \gcd(a, n) = 1\}$; thus $\varphi(1) = 1$. OEIS A000010 records these values.

Theorem 1 (Euler's formula)**.** *For every positive integer n,*

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right) = \prod_{p^e \| n} p^{e-1}(p-1), \tag{1}$$

where the first product runs over distinct prime divisors and the second runs over exact prime powers in the factorization. Both products are 1 for $n = 1$.

Proof. Among the integers $1, \ldots, n$, a number is not relatively prime to n exactly when it is divisible by at least one distinct prime factor of n. Inclusion–exclusion counts those divisible by none as

$$\sum_{S \subseteq \{p : p \mid n\}} (-1)^{|S|} \frac{n}{\prod_{p \in S} p} = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right),$$

since each product of selected primes divides n. Writing $n = \prod_{p^e \| n} p^e$ gives the second expression. □

Theorem 2 (Divisor identity and inversion)**.** *For all $n \geq 1$,*

$$\sum_{d \mid n} \varphi(d) = n, \qquad \varphi(n) = \sum_{d \mid n} \mu(d) \frac{n}{d}. \tag{2}$$

Here μ is the Möbius function.

Proof. Partition $1, \ldots, n$ according to $g = \gcd(k, n)$. If $g \mid n$, the numbers with this gcd are exactly $k = gu$ where $1 \leq u \leq n/g$ and $\gcd(u, n/g) = 1$. There are $\varphi(n/g)$ of them. Summing over $g \mid n$ gives the first identity. Apply Möbius inversion to $n = \sum_{d \mid n} \varphi(d)$ to obtain the second. Alternatively, expand the product in (1); all squarefree divisors contribute $\mu(d)n/d$, while nonsquarefree divisors contribute zero. □

2 Exact implications for primality

Proposition 1 (Totient criterion)**.** *For every integer $n \geq 2$,*

$$\boxed{n \text{ is prime} \iff \varphi(n) = n - 1.}$$

Moreover, $\varphi(n)$ is even for every $n > 2$.

Proof. If n is prime, every number $1, \ldots, n-1$ is coprime to n, so $\varphi(n) = n - 1$. If n is composite, it has a prime divisor p with $1 < p < n$, hence p is an element of $1, \ldots, n-1$ not counted by $\varphi(n)$, giving $\varphi(n) \leq n - 2$.

For parity, whenever a is a unit modulo n, so is $n - a$. For $n > 2$, these are distinct: equality would force $2a = n$, which yields $\gcd(a, n) = a = n/2 > 1$ when integral. Thus the units pair off, and their number is even. □

3 Numerical check and scope

For $n = 12 = 2^2 \cdot 3$, the product formula gives $\varphi(12) = 12(1 - 1/2)(1 - 1/3) = 4$, and

$$\sum_{d \mid 12} \varphi(d) = \varphi(1) + \varphi(2) + \varphi(3) + \varphi(4) + \varphi(6) + \varphi(12) = 1 + 1 + 2 + 2 + 2 + 4 = 12.$$

The first twenty values were compared against the OEIS entry. The equivalence $\varphi(n) = n - 1$ is a characterization, not a claim of a faster primality test: computing $\varphi(n)$ may be difficult without factorization. All assertions here are classical; none resolves an open problem.

References

[1] OEIS Foundation, *A000010: Euler totient function $\varphi(n)$*, https://oeis.org/A000010 (accessed November 13, 2026).

[2] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976 (totients, arithmetic functions and Möbius inversion).